

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check that the cylinder locking nut is tight
Figure 12.2-34 Example of a technician checking the cylinder locknut	
Monthly	Check that the rod "fork" is correctly attached
Figure 12.2-35 Example of a technician checking that the circlips are in place and the mechanism is secure	



Caution !

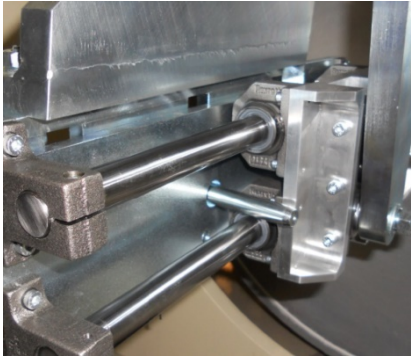
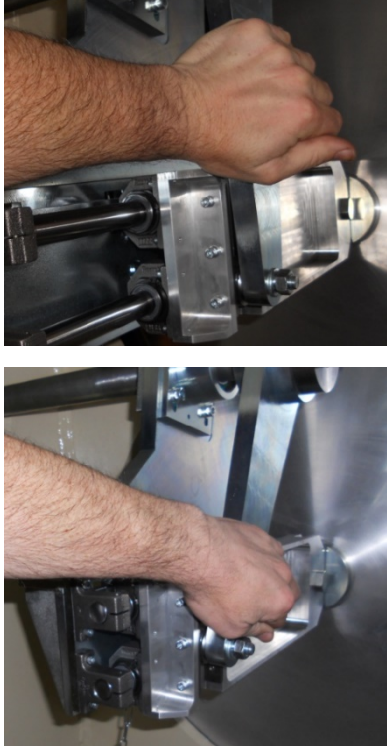
Ensure that the air supply has been removed before carrying out further actions as described below

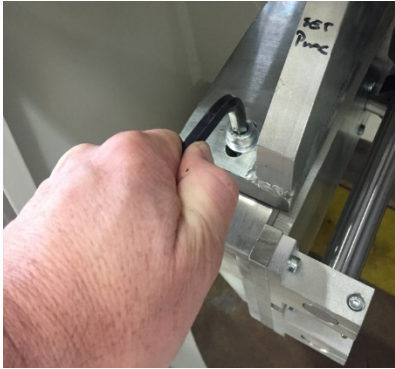


Caution !

Let the diffusion pumps cool down to ambient temperature before accessing the inside components

Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	With the machine powered down and locked out, vent the pump to atmosphere through the hand valve where the pirani gauge is located. Remove the inspection/maintenance cover from the outside.
Figure 12.2-36 Location of the hand valve with the access panel removed	
Bi-Annual	Access the internal mechanism of the valve to make the following checks. The air supply should be locked out and the pump brought to atmosphere. Disconnect the air supply from both sides of the cylinder
Figure 12.2-37 Example of the external cylinder with air the lines disconnected	

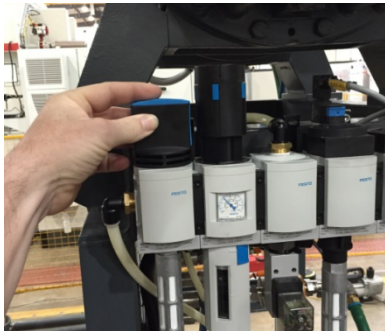
Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Check that the locking pin is available and can be inserted correctly to support the valve to allow for seal cleaning, etc.
Figure 12.2-38 Example of a technician checking that locking pin can be inserted	
Bi-Annual	Check the correct tightness of the external and internal ETP bushes, using a torque wrench. The value of torque is etched on the side of the bush
Figure 12.2-39 The location of the internal ETP bush	
Bi-Annual	Check the movement of the assembly from fully open to fully closed. Visually check for any indication of any wear
Figure 12.2-40 Example of a technician checking the movement of the assembly	

Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Check that all assembly bolts/fasteners are correctly tightened to the recommended torque level
Figure 12.2-41 Example of a technician checking the fastener's for tightness	
Bi-Annual	Check o-ring and sealing surface of inspection panel
Figure 12.2-42 Example of a technician checking the o-ring and sealing surfaces on the inspection panel	

Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Inspect the valve seat with the manual override on the controlling spool to check o-ring condition and mating surfaces. Replace o-rings as required
Figure 12.2-43 Example of a technician checking the condition of the o-ring and sealing surface	

12.2.6 Pneumatic (Air) Systems

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Check all air lines for leaks.

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check the operation of the pressure regulator and isolation valve
Figure 12.2-44 Example of a technician checking the operation of the air supply regulator and isolation valve	
Monthly	Check the operation of the air healthy monitor switch by reducing the pressure using the regulator valve and checking that it operates at approximately 3 bar
Monthly	Visually check for any indication of damage to other pneumatically driven components

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Check the condition of control cabling to the pneumatic devices

Maintenance Requirement	Equipment & Task to Be Performed
10 Years	Change to flexible pipelines

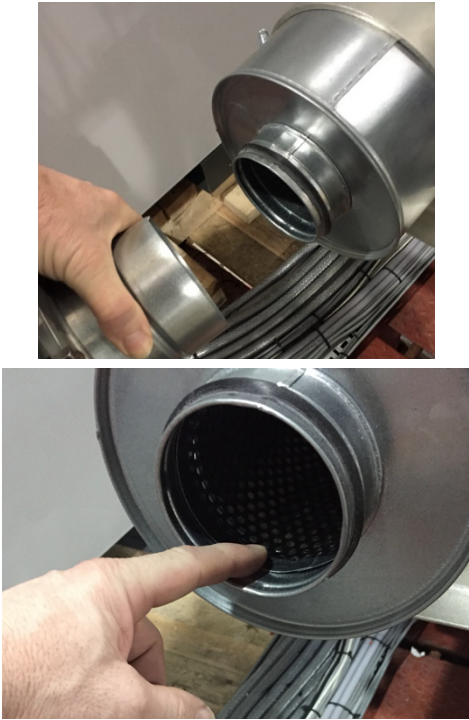
12.2.7 Vacuum Gauges

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check the condition of all the vacuum gauges and their wiring for physical damage.
Monthly	Check the operation of the isolation valves between the vacuum lines and the gauges, where appropriate.

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Compare and calibrate MKS pirani gauges against a known standard gauge. See chapter 11.3 Vacuum Gauge Maintenance on page 532.
Figure 12.2-45 MKS pirani gauge	
Quarterly	Clean the penning gauge as per the vendor's manual's instructions.
Figure 12.2-46 Example of a technician unscrewing the electronic body from the penning gauge tube	

Maintenance Requirement	Equipment & Task to Be Performed
5 Years	Replace all vacuum gauges

12.2.8 Chamber Vent Valves

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check the condition of any filters installed Clean or replace as required. Monitor the time required for the chamber to be vented to atmosphere, this will indicate the replacement timing of the internal filter media.
Figure 12.2-47 External vent valve	
Figure 12.2-48 Example of a technician checking the condition of the filters on the external section of the vent valve	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Clean the internal deflection plate inside the chamber.
Figure 12.2-49 Example of technician checking the internal baffle plate	

12.2.9 Vacuum System O-rings

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Check main faceplate to chamber o-ring for damage. Repair or replace as necessary.
Figure 12.2-50 Example of a technician checking the main chamber sealing area	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Wipe down debris from main faceplate to chamber o-ring and regrease.

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Open the VV7 valve plates, inspect, wipe down any debris from the o-ring and regrease.
Figure 12.2-51 Example of a technician checking the VV7 valve seat	

Maintenance Requirement	Equipment & Task to Be Performed
Annual	Replace the o-ring/cord in the high vacuum valve seats (VV7).
Figure 12.2-52 VV7 valve seat	
Annual	Grease all inspection port o-rings, where practical.

Maintenance Requirement	Equipment & Task to Be Performed
5 Years	Replace o-rings on machine as components are removed for maintenance, "as they can potentially dry out and crack, therefore creating leaks.

12.2.10 Vacuum Pipeline Rubber Isolation Bellows

The rubber bellows installed in the pipelines can be removed to allow access to remove sections of the pipework. The hose clamps should be loosened. Then the bellows pulled away from the pipe-work that is to be removed. Before re-installing check the pipework for any rough edges and clean them up. There should be a maximum gap of 12mm between the two edges of the pipe. Add vacuum grease to the inner surface of the seal and gently push the bellows back over the joint. Then re-tighten the hose clamps.



Figure 12.2-53 Gap between vacuum pipes

Maintenance Requirement		Equipment & Task to Be Performed	
Weekly		Visually check for wear or damage to the bellows. Replace as necessary.	
Figure 12.2-54 Example of a technician checking the rubber bellows			
Weekly		Check for excess movement - this can lead to air leaks	

Maintenance Requirement	Equipment & Task to Be Performed		
Monthly	Check the tightness of the hose clamps. There should be two on either side of the joint. The clamp section should be 180 degrees from each other, to stop bellows being made oval by clamping force.		

12.2.11 Cryo-Generator – Brookes Maxcool Mk 2

(refer to the vendor manual for details)

Brookes recommends that the user inspect the system at daily, weekly, monthly and quarterly intervals. Brookes recommends that every 12 months a preventive maintenance service visit is performed by Brookes authorised service agent. They also recommend the use of the record sheets in their manuals for monitoring the unit's performance.

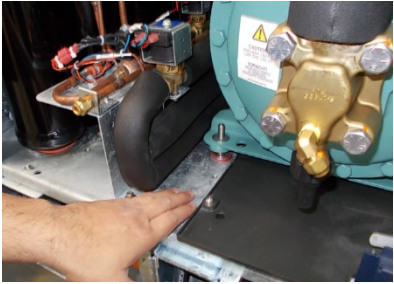
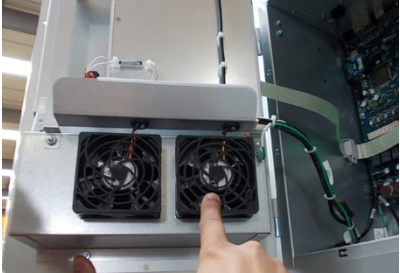
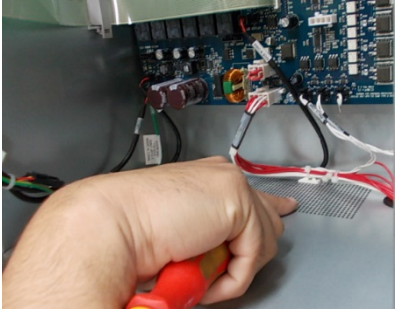
Maintenance Requirement	Equipment & Task to Be Performed
Daily	Ensure the unit is functioning as expected

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check Temperatures (during process), as displayed on the external LCD panel. Record this information for comparison purposes.
Figure 12.2-55 Example of a technician checking the operating parameters on the HMI display on the cryogenerator unit	
Weekly	Check suction and discharge refrigerant gas pressures (during process) as displayed on the external LCD panel. Record this information for comparison purposes.

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check the time for the unit to cool down the cryogenic coil and record this information, for comparison purposes.

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Check refrigeration system balance pressure, as described in the vendor manual, using the HMI interface and record this information, for comparison purposes.
Quarterly	Visually verify LCD display is not missing characters
Quarterly	Visually verify HMI keypad / Remote Control functions properly

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Visually verify that the interconnection pipework insulation is intact. Make any repairs as required.
Figure 12.2-56 Example of a technician checking the condition of the insulation	
Quarterly	Visually verify oil level in compressor is appropriate. Refer to vendor manual
Figure 12.2-57 Example of a technician checking the oil level of the compressor	
Maintenance Requirement	Equipment & Task to Be Performed
Annual	Replace valve and valve components as necessary
Annual	Review event files for errors and backup system data (future capability)
Annual	Check cool down cycle time, pressure, and temperature, using the HMI interface and record this information, for comparison purposes.
Annual	Check defrost cycle time, pressure, and temperature, using the HMI interface and record this information, for comparison purposes.
Annual	Verify water flow and temperatures by measuring the differential between the water inlet and outlet temperature, while cooling down the cryogenic coil, and record this information, for comparison purposes.
Figure 12.2-58 Example of a technician checking the inlet and outlet water temperatures	

Maintenance Requirement	Equipment & Task to Be Performed
Annual	Inspect frame, compressor, and components for wear
Figure 12.2-59 Example of a technician visually checking for corrosion internally	
Annual	Re-insulate pipelines (as appropriate)
Annual	Perform refrigerant “leak check” on fittings within the unit and vacuum feed-throughs, using a suitable electronic detector
Figure 12.2-60 Example of a technician checking a refrigerant leak check on the interconnection pipelines	
Annual	Clean electronic control box
Figure 12.2-61 Example of a technician cleaning the electrical enclosure	
Annual	Inspect all wiring harnesses and electrical connections
Figure 12.2-62 Example of a technician checking electrical harness terminations	

12.2.12 Vacuum System Performance

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Visually check for indication of water leaks within the chamber.

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Vacuum leak check the chamber, pumps and pipe work for air leaks using a helium leak detector

Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annual	Check the ultimate pressure of the system by evacuating and pumping the chamber overnight. Do not leave rolls of product in the machine. The target value is less than $2e^{-5}$ mbar

	Note !
	Establish a vacuum system performance baseline that can be used to compare the vacuum pumping system performance against at later stages of the machine life cycle

12.2.13 Vacuum Chamber

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Clean and lubricate the winding mechanism joint face o-ring with a suitable grease.
Weekly	Check all joint faces for damage, and make repairs as required
Figure 12.2-63 Example of a technician checking the O-ring and surface area of the winding cart	
Weekly	Clean the interior walls as per the instructions given in the operation manual.

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check all the spool valves for correct operation
Figure 12.2-64 Example of a technician checking the chamber joint mating surface	

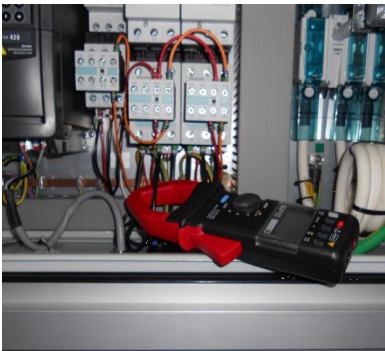
12.2.14 GRE Chiller Unit (48KW- 96KW Models)

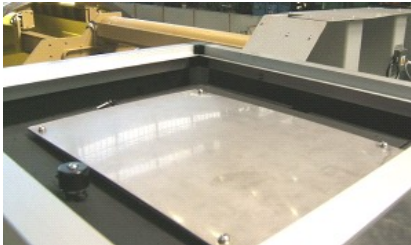
Maintenance Requirement	Equipment & Task to Be Performed
Daily	Check externally for water leaks
Figure 12.2-65 Chiller with the access panel removed	
Daily	Check for unusual operating noises

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check externally for water leaks
Figure 12.2-66 Example of a technician checking for water leaks	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check that external panel location fasteners are fully engaged.
Figure 12.2-67 Check the panel external fasteners are secure	

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Check suction and discharge pressures are correct on Interface screen

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Check sight glass for dryness and bubbles to verify that there are no refrigerant system leaks
Figure 12.2-68 Example of a technician checking the refrigerant sight glass	
Quarterly	Check the compressor oil levels and refer to the vendor manual for advice on the level required.
Figure 12.2-69 Example of a technician checking the oil level indicator located between the compressors	
Quarterly	Measure the feed and return water temperatures during a pull down sequence, when the machine has been placed in the "Vacuum mode" to record the difference between them at a set chiller temperature. This will give indication of potential issues with the water filter or potential blockage in the condenser.
Quarterly	Check correct operation of the pump and flow switch, by monitoring and recording the values indicated on the HMI. Record these for future reference.
Quarterly	Check the current being drawn by the units components, such as pumps, heaters and compressors. This information will give indication of potential problems
Figure 12.2-70 Example of a technician checking the pump and compressor motor currents	

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Clean the water supply condenser water filter
Figure 12.2-71 Example of a technician removing the cover to clean the condenser water filter	
Quarterly	Check that internal pipework clamps are secure
Figure 12.2-72 Example of a technician checking that the pipe work clamps are secure	
Quarterly	Check that compressor mounts are in good condition and are tight
Figure 12.2-73 Example of a technician checking that the compressor mounts are in good condition and are tight	
Quarterly	Visually check the pressure relief valves for signs on operation (there should not normally be any)
Quarterly	Check the coolant concentration (Glycol). Changes of weakening or strengthening fluid density will highlight other possible issues. Take a sample from the hot tank.
Figure 12.2-74 Example of a technician checking the coolant concentration by accessing the hot tank from the top of the unit.	

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Check that the temperature measuring devices are in good contact with the source of contact
Figure 12.2-75 Example of a technician checking the temperature measuring devices	
Quarterly	Check that the compressor band heater is operational. If not operational, there could be issues with liquid refrigerant in the compressors. These heaters are operational when the compressor is not operational.
Figure 12.2-76 Example of a technician touching the compressor heater to check that it is operational	
Quarterly	Check the glycol mixture for strength and for contamination. Modify by adding/removing glycol/water and add chemicals to control biocides. Remove and clean the glycol process line strainer.
Figure 12.2-77 Example of a technician checking the glycol process line strainer	

Maintenance Requirement	Equipment & Task to Be Performed
	The filter is built into a ball valve, that when placed in the horizontal position allows the user to remove the access cap and clean out the mesh. Refer to the Marflow Ball Filter cleaning procedure, listed under GRE in the vendor manuals section of the documentation supplied.
Figure 12.2-78 Position of the process line filter	 
Figure 12.2-79 Guidance label on the insulation covering the strainer access	

	Caution !
	There will be some spillage of coolant, which should be cleaned away. When reinserting the strainer and cap, always use a suitable grease, to stop the stainless steel components “picking up”

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Check that all the small capillary tubing is in a good condition, between the refrigerant lines and monitoring sensors/switches.
Quarterly	Check that the pump and compressor motor currents do not exceed the rating on the motor plates under normal running conditions.
Quarterly	Check the operation of the flow switch by restricting the flow at the outlet valve, by closing it.
Quarterly	Check pipe work insulation and make repairs as necessary.

Maintenance Requirement		Equipment & Task to Be Performed	
Bi-Annually		Check all electrical connections are tight.	
Figure 12.2-80 Example of a technician checking the electrical connections			

Maintenance Requirement	Equipment & Task to Be Performed		
Annually	The chiller manufacturer recommends an annual full service by a qualified refrigeration engineer.		

12.2.15 Evaporation Source Viewing Flap & Viewing Assembly

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Visually check for potential water leaks and mechanical damage
Daily	Visually check for mechanical damage
Daily	Manually spin rotary section to check that it is not binding with other components.

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Grease bearings at either end of the assembly
Figure 12.2-81 Example of a technician greasing the assembly bearings	
Weekly	Visually check that the coupling between the rotary strobe and the motor is engaged correctly with a 3mm space
Weekly	Grease the rotary pivot points at either end of the assembly
Figure 12.2-82 Example of a technician greasing the rotary pivot point	
Weekly	Check for correct operation of flap using spanner provided.

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check that there is no binding between the components
Figure 12.2-83 Example of a technician visually checking the ends of the flap for binding	
Weekly	Visually check rotary vacuum leadthrough, externally for indication of oil loss. Investigate
Figure 12.2-84 Example of a technician visually checking the leadthrough gravity bottle	
Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check Universal joint and splines for correct movement
Figure 12.2-85 Example of a technician visually checking the spline for clear movement	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check Universal joint and splines for correct movement
Figure 12.2-86 Example of a technician visually checking the universal joint cover	

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Remove the motor and spin test the strobe cylinder to check the condition of the bearings

Maintenance Requirement	Equipment & Task to Be Performed
Annual	Change assembly bearings

Maintenance Requirement	Equipment & Task to Be Performed
2 Years	Change rubber segment in drive coupling
2 Years	Replace the bushes, where rotation takes place.

12.2.16 MKS Mass Flow Controllers

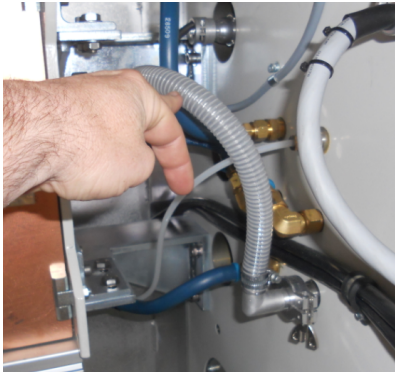
  	Warnings ! Beware that leakage of process gases can create a localized environment where an asphyxiation hazard is created. Escaping combustible process gases can create a local atmosphere in which an explosion can occur. Be aware that these components are under a pressurised situation internally.
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Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Visually check unit and electrical control cabling for physical damage and replace as necessary.
Figure 12.2-87 Example of a technician checking for damage to MKS gas controller	
Monthly	Check that the “shut off” isolation valve is operating correctly, by removing the electrical connection, and operating the gas flow from the Plasma HMI screen. An alarm should be shown.
Figure 12.2-88 Example of a technician removing the electrical connection solenoid	

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Visually check the condition of the gas line from the gas supply/bottle connection through the cable coiler to the flow controller. Use a liquid gas leak detection substance (rated for the application), if leakage suspected.
Figure 12.2-89 Example of a technician checking for gas leaks using a suitable leak detection liquid	 A close-up photograph showing a person's hand holding a white bottle with a blue label that says 'COOLANT' and 'MFC'. The person is applying the liquid from the bottle to a metal gas line connection. The connection is part of a larger piece of equipment, possibly a flow controller, which has various pipes and valves. The background is slightly blurred, focusing on the application of the leak detection liquid.

12.2.17 MKS Baratron Vacuum Gauge

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Visually check for physical damage and replace as necessary
Monthly	Check the cable connections to the unit
Figure 12.2-90 Example of a technician checking for damage to MKS Baratron gauge	

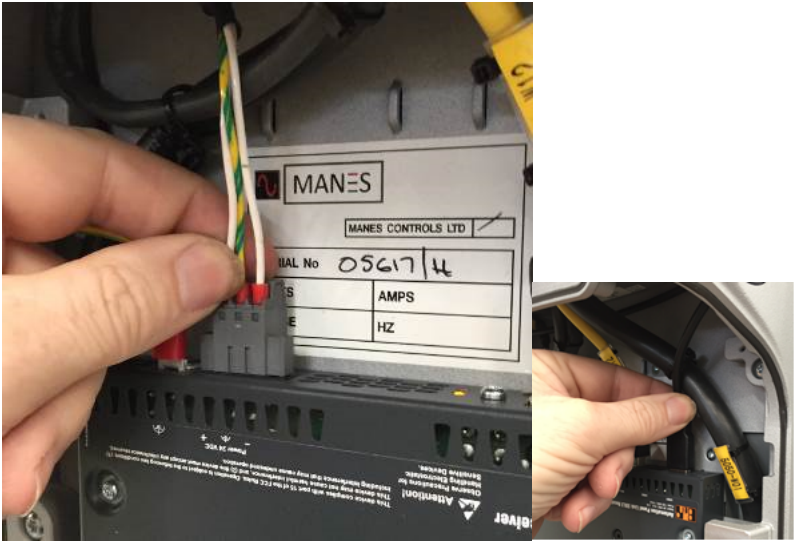
Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Check that the shut off valve is operating correctly, by removing the electrical connection and trying to flow gas.
Quarterly	Visually check the condition of the internal pipework from faceplate to the plasma treater
Figure 12.2-91 Example of a technician checking for damage to Baratron internal pipework	

Maintenance Requirement	Equipment & Task to Be Performed
Annually	Check calibration of gauge to instructions in vendor manual

12.2.18 Operator HMI Unit

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check the calibration of the touch screen and reset using the on board ELO software if required.
Figure 12.2-92 Example of a technician checking the movement of the operator interface unit	

Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annually	Open the rear panel and check that there is no damage to the cables as they pass out of the panel through the connecting arm
Figure 12.2-93 Example of a technician checking the cabling passing out of the unit	

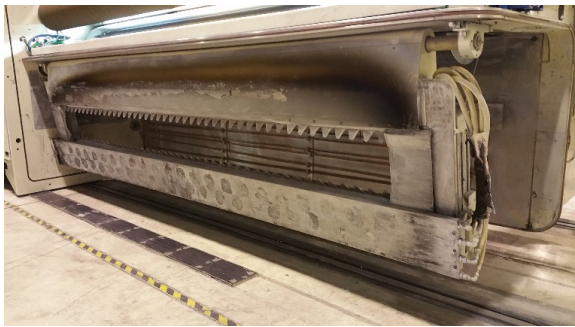
Maintenance Requirement	Equipment & Task to Be Performed
Bi-Annually	Open the rear panel and check that there is no indication of contamination of any of the components, Use a vacuum cleaner to remove any contamination
Figure 12.2-94 Example of a technician checking the connections on the switches	
Figure 12.2-95 Example of a technician checking the cable terminations	

12.2.19 Coating Shield & Shutter



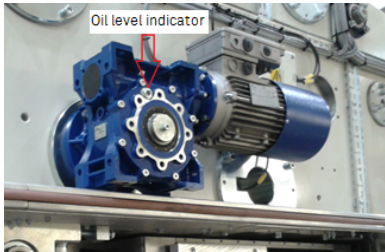
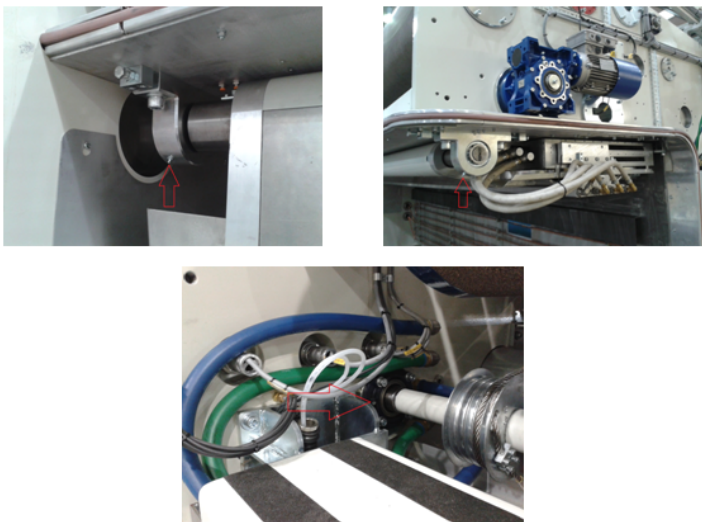
Warning!

Make sure that the incoming energy source is switched off and secured by a padlock (lock out tag system) when any maintenance work is being carried out.

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Grease the shutter movement mechanism bearings.
Figure 12.2-96 Example of a technician greasing the bearings on the shutter mechanism	
Weekly	Visually check the drive tracks for excessive wear, if worn schedule a replacement
Weekly	Check the operation of the movement proximity switches.
Weekly	Check the shutter for mechanical damage.
Figure 12.2-97 Check the shutter for physical damage	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Visually, check the shutter water supply hoses for indication of cracking or physical wear.
Figure 12.2-98 Example of a technician checking the water hoses	
Figure 12.2-99 Example of a technician checking the water hoses	
Figure 12.2-100 Example of a technician checking the water hoses	
Figure 12.2-101 Example of a technician checking the flexible cables for indications of wear or trapment.	

Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2-102 Example of a technician checking the flexible cables for indications of wear or trapment.	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check oil level in winch gearbox – top up as necessary with Anderol 555 oil.
Figure 12.2-103 Oil level indicator on the winch gearbox	
Monthly	Apply grease type A (ie, Mobil SHC32 or equivalent) to the pivot bearing housings and winch shaft bearing as shown.
Figure 12.2-104 Grease points on the pivot bearing housing and winch shaft bearing	

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Remove access covers and clear out any debris in the motor enclosures.